Technical Traceability Appendix

Independent Model Validation Sample Review

Demonstration Package - Independent Validation Review

Independent Model Validation Review

Technical Traceability Appendix

Lifecycle State: FROZEN Evidence Authority: Frozen Step 2.2 Controlled Evidence Set

A. Canonical Quantitative Metrics

Evidence ID	Metric / Context	Value	Frozen Source	Source Location
EVD-370	AUC — TEST — BENCHMARK = 0.763472148601427	0.763472148601427	phase-3-independent-validation/3.6-performance-benchmark-and-robustness-testing/evidence/independent-testing/independent-baseline-performance-reconciliation.csv	Line 7
EVD-371	AUC — TEST — CHALLENGER = 0.781502340482797	0.781502340482797	phase-3-independent-validation/3.5-implementation-verification/evidence/challenger-metric-reconciliation.csv	Line 12
EVD-372	AUC — TEST — PRIMARY = 0.763	0.763	phase-3-independent-validation/3.2-documentation-and-evidence-review/docs/primary-documentation-review-packet.md	Line 1197
EVD-373	AUC — TEST — UNSPECIFIED = 0.763472148601427	0.763472148601427	phase-3-independent-validation/3.5-implementation-verification/evidence/champion-metric-reconciliation.csv	Line 21
EVD-374	BRIER — TEST — BENCHMARK = 0.137875813164177	0.137875813164177	phase-3-independent-validation/3.6-performance-benchmark-and-robustness-testing/evidence/independent-testing/independent-baseline-performance-reconciliation.csv	Line 10
EVD-375	BRIER — TEST — CHALLENGER = 0.135260187115186	0.135260187115186	phase-3-independent-validation/3.5-implementation-verification/evidence/challenger-metric-reconciliation.csv	Line 15
EVD-376	GINI — TEST — BENCHMARK = 0.526944297202854	0.526944297202854	phase-3-independent-validation/3.6-performance-benchmark-and-robustness-testing/evidence/independent-testing/independent-baseline-performance-reconciliation.csv	Line 8
EVD-377	GINI — TEST — CHALLENGER = 0.563004680965595	0.563004680965595	phase-3-independent-validation/3.5-implementation-verification/evidence/challenger-metric-reconciliation.csv	Line 13
EVD-378	GINI — TEST — UNSPECIFIED = 0.526944297202854	0.526944297202854	phase-3-independent-validation/3.5-implementation-verification/evidence/champion-metric-	Line 22

Evidence ID	Metric / Context	Value	Frozen Source	Source Location
			reconciliation.csv	
EVD-379	KS — TEST — BENCHMARK = 0.393988393295287	0.393988393295287	phase-3-independent-validation/3.6-performance-benchmark-and-robustness-testing/evidence/independent-testing/independent-baseline-performance-reconciliation.csv	Line 9
EVD-380	KS — TEST — CHALLENGER = 0.419950198925315	0.419950198925315	phase-3-independent-validation/3.5-implementation-verification/evidence/challenger-metric-reconciliation.csv	Line 14
EVD-381	LOG_LOSS — TEST — BENCHMARK = 0.440715765175037	0.440715765175037	phase-3-independent-validation/3.6-performance-benchmark-and-robustness-testing/evidence/independent-testing/independent-baseline-performance-reconciliation.csv	Line 11
EVD-382	LOG_LOSS — TEST — CHALLENGER = 0.430431310056478	0.430431310056478	phase-3-independent-validation/3.5-implementation-verification/evidence/challenger-metric-reconciliation.csv	Line 16

B. Independent Validator Judgments

Evidence ID	Judgment ID	Judgment	Frozen Source	Source Location
EVD-342	IQ-001	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 2
EVD-343	IQ-002	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 3
EVD-344	IQ-003	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 4
EVD-345	IQ-004	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 5
EVD-346	IQ-005	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 6
EVD-347	IQ-006	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/	Line 7

Evidence ID	Judgment ID	Judgment	Frozen Source	Source Location
			implementation-review-questions.csv	
EVD-348	IQ-007	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 8
EVD-349	IQ-008	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 9
EVD-350	IQ-009	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 10
EVD-351	IQ-010	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 11
EVD-352	IQ-011	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 12
EVD-353	IQ-012	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 13
EVD-354	IQ-013	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 14
EVD-355	IQ-014	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 15
EVD-356	IQ-015	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 16
EVD-357	IQ-016	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 17
EVD-358	IQ-017	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 18

Evidence ID	Judgment ID	Judgment	Frozen Source	Source Location
EVD-359	IQ-018	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 19
EVD-360	IQ-019	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 20
EVD-361	IQ-020	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 21
EVD-362	IQ-021	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 22
EVD-363	IQ-022	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 23
EVD-364	IQ-023	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 24
EVD-365	IQ-024	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 25
EVD-366	IQ-025	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 26
EVD-367	IQ-026	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 27
EVD-368	IQ-027	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 28

C. Complete Client-Facing Evidence Register

Evidence ID	Type	Controlled Statement	Frozen Value	Frozen Source	Source Location
EVD-340	Fact	Formal finding candidates: 0	0	phase-3-independent-	Line 1525

Evidence ID	Type	Controlled Statement	Frozen Value	Frozen Source	Source Location
				validation/3.5-implementation-verification/automation/finalize-substantive-implementation-judgment.py	
EVD-341	Fact	Implementation observations: 0	0	phase-3-independent-validation/3.5-implementation-verification/automation/finalize-substantive-implementation-judgment.py	Line 1523
EVD-342	Judgment	IQ-001 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 2
EVD-343	Judgment	IQ-002 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 3
EVD-344	Judgment	IQ-003 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 4
EVD-345	Judgment	IQ-004 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 5
EVD-346	Judgment	IQ-005 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 6
EVD-347	Judgment	IQ-006 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 7
EVD-348	Judgment	IQ-007 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-	Line 8

Evidence ID	Type	Controlled Statement	Frozen Value	Frozen Source	Source Location
				implementation-verification/config/implementation-review-questions.csv	
EVD-349	Judgment	IQ-008 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 9
EVD-350	Judgment	IQ-009 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 10
EVD-351	Judgment	IQ-010 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 11
EVD-352	Judgment	IQ-011 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 12
EVD-353	Judgment	IQ-012 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 13
EVD-354	Judgment	IQ-013 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 14
EVD-355	Judgment	IQ-014 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 15
EVD-356	Judgment	IQ-015 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-	Line 16

Evidence ID	Type	Controlled Statement	Frozen Value	Frozen Source	Source Location
				review-questions.csv	
EVD-357	Judgment	IQ-016 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 17
EVD-358	Judgment	IQ-017 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 18
EVD-359	Judgment	IQ-018 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 19
EVD-360	Judgment	IQ-019 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 20
EVD-361	Judgment	IQ-020 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 21
EVD-362	Judgment	IQ-021 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 22
EVD-363	Judgment	IQ-022 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 23
EVD-364	Judgment	IQ-023 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 24
EVD-365	Judgment	IQ-024 —	SATISFACTORY	phase-3-	Line 25

Evidence ID	Type	Controlled Statement	Frozen Value	Frozen Source	Source Location
		SATISFACTORY		independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	
EVD-366	Judgment	IQ-025 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 26
EVD-367	Judgment	IQ-026 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 27
EVD-368	Judgment	IQ-027 — SATISFACTORY	SATISFACTORY	phase-3-independent-validation/3.5-implementation-verification/config/implementation-review-questions.csv	Line 28
EVD-369	Judgment	- PKG-003 â€” Final Validation Disposition â€” phase-3-independent-validation/3.9-final-validation-disposition-and-package/evidence/final-validation-disposition.csv		phase-3-independent-validation/3.9-final-validation-disposition-and-package/final-package/README.md	Line 16
EVD-370	Metric	AUC — TEST — BENCHMARK = 0.763472148601427	0.763472148601427	phase-3-independent-validation/3.6-performance-benchmark-and-robustness-testing/evidence/independent-testing/independent-baseline-performance-reconciliation.csv	Line 7
EVD-371	Metric	AUC — TEST — CHALLENGER = 0.781502340482797	0.781502340482797	phase-3-independent-validation/3.5-implementation-verification/evidence/challenger-metric-reconciliation.csv	Line 12
EVD-372	Metric	AUC — TEST — PRIMARY = 0.763	0.763	phase-3-independent-validation/3.2-documentation-and-evidence-review/docs/primary-	Line 1197

Evidence ID	Type	Controlled Statement	Frozen Value	Frozen Source	Source Location
				documentation-review-packet.md	
EVD-373	Metric	AUC — TEST — UNSPECIFIED = 0.763472148601427	0.763472148601427	phase-3-independent-validation/3.5-implementation-verification/evidence/champion-metric-reconciliation.csv	Line 21
EVD-374	Metric	BRIER — TEST — BENCHMARK = 0.137875813164177	0.137875813164177	phase-3-independent-validation/3.6-performance-benchmark-and-robustness-testing/evidence/independent-testing/independent-baseline-performance-reconciliation.csv	Line 10
EVD-375	Metric	BRIER — TEST — CHALLENGER = 0.135260187115186	0.135260187115186	phase-3-independent-validation/3.5-implementation-verification/evidence/challenger-metric-reconciliation.csv	Line 15
EVD-376	Metric	GINI — TEST — BENCHMARK = 0.526944297202854	0.526944297202854	phase-3-independent-validation/3.6-performance-benchmark-and-robustness-testing/evidence/independent-testing/independent-baseline-performance-reconciliation.csv	Line 8
EVD-377	Metric	GINI — TEST — CHALLENGER = 0.563004680965595	0.563004680965595	phase-3-independent-validation/3.5-implementation-verification/evidence/challenger-metric-reconciliation.csv	Line 13
EVD-378	Metric	GINI — TEST — UNSPECIFIED = 0.526944297202854	0.526944297202854	phase-3-independent-validation/3.5-implementation-verification/evidence/champion-metric-reconciliation.csv	Line 22
EVD-379	Metric	KS — TEST — BENCHMARK = 0.393988393295287	0.393988393295287	phase-3-independent-validation/3.6-performance-benchmark-and-robustness-testing/evidence/independent-testing/independent-baseline-	Line 9

Evidence ID	Type	Controlled Statement	Frozen Value	Frozen Source	Source Location
				performance-reconciliation.csv	
EVD-380	Metric	KS — TEST — CHALLENGER = 0.419950198925315	0.419950198925315	phase-3-independent-validation/3.5-implementation-verification/evidence/challenger-metric-reconciliation.csv	Line 14
EVD-381	Metric	LOG_LOSS — TEST — BENCHMARK = 0.440715765175037	0.440715765175037	phase-3-independent-validation/3.6-performance-benchmark-and-robustness-testing/evidence/independent-testing/independent-baseline-performance-reconciliation.csv	Line 11
EVD-382	Metric	LOG_LOSS — TEST — CHALLENGER = 0.430431310056478	0.430431310056478	phase-3-independent-validation/3.5-implementation-verification/evidence/challenger-metric-reconciliation.csv	Line 16

D. Controlled Final Validation Disposition

- PKG-003 â€” Final Validation Disposition â€” phase-3-independent-validation/3.9-final-validation-disposition-and-package/evidence/final-validation-disposition.csv